

LESSON 3-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Graphing Ratios on the Coordinate Plane**

Maryland Standard 6.RP.A.3a • Learning target – I can turn a ratio into ordered pairs and plot them on a coordinate plane.

REMEMBER

- Write each pair from a ratio table as an **ordered pair** (**x, y**). The first quantity is x.
- Points for equivalent ratios lie on a **straight line** through **(0, 0)**.

WORKED EXAMPLE

A pack holds 4 juice boxes. Let x = packs and y = juice boxes. Write ordered pairs for 1, 2, and 3 packs.

Step 1 Table: 1 pack, 4 boxes; 2 packs, 8 boxes; 3 packs, 12 boxes

Step 2 Ordered pairs: (1, 4), (2, 8), (3, 12)

Answer **(0, 0), (1, 4), (2, 8), (3, 12)** lie on one line.

WORD BANK

ordered pair – (x, y): x across, then y up
origin – the point (0, 0)

x-axis – the line that goes across
y-axis – the line that goes up

PRACTICE

show your work • label units

1. Each bag holds 3 oranges. Complete the table. Then write each column as an ordered pair.

Step 1 Multiply each number of bags by 3.

Step 2 Write each column as (bags, oranges).

Bags (x)	1	2	3	4
Oranges (y)	3			
(x, y)	(1, 3)			

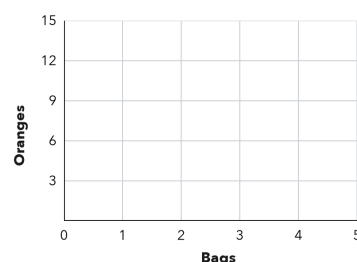
2. Plot the ordered pairs from Exercise 1. Draw a line through the points. Does the line pass through (0, 0)? What does (0, 0) mean?

Step 1 Start at (0, 0). Go across to the x-number.

Step 2 Then go up to the y-number and draw a dot.

Step 3 Use a ruler to connect the dots.

(0, 0) means _____



3. A muffin batch uses 2 cups of flour (x = batches, y = cups). Write the ordered pair for:
- a. 7 batches _____ b. 18 cups of flour _____

4. Garden Rows. Each **row** has **5 plants**. Let x = rows and y = plants.

Step 1 Multiply the rows by 5 to get y .

Step 2 Write the point as (rows, plants).

a. Write the ordered pair for 8 rows. _____

b. Write the ordered pair for 0 rows. What does it mean? _____

5. Same Line? Do the points (1, 5), (2, 10), and (3, 14) all lie on one straight line through (0, 0)?

☐ Yes

☐ No

6. Find the Error. Tickets cost \$2 each. On a graph with x = tickets and y = cost, Ruby plotted (12, 6) for 6 tickets. Explain her mistake and write the correct point.

Frame Ruby's mistake is _____. The correct point is (____, _____) because _____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Make a table for something that comes in packs at home. Say two of its ordered pairs.

LESSON 3-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Interpreting Points on Ratio Graphs**

Maryland Standard 6.RP.A.3a • Learning target – I can explain what a point on a ratio graph means in the situation it describes.

REMEMBER

- Each point (x, y) is a pair of amounts. Say **both** with units.
- The point **$(1, y)$** shows the **unit rate**: $y \div x$ for any point. $(0, 0)$ means zero of both.

WORKED EXAMPLE

A graph of hours worked (x) and dollars earned (y) passes through $(3, 45)$. What does the point mean?

Step 1 $(3, 45)$: 3 hours of work earns \$45

Step 2 Unit rate: $45 \div 3 = 15$, so $(1, 15)$ is on the line

Answer \$15 per hour

WORD BANK

unit rate – the amount for 1

on the line – $y \div x$ gives the same unit rate

point – (x, y) : x of one thing goes with y of another

PRACTICE

show your work • label units

1. The graph shows time (x , hours) and distance (y , miles) for a cyclist.

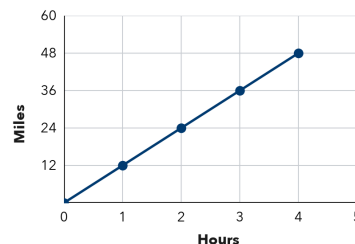
Step 1 Read the x -axis label and the y -axis label.

Step 2 Find the point where $x = 1$. Its y is the unit rate.

a. What does the point $(2, 24)$ mean? _____

b. What is the unit rate? _____ miles per hour

c. At this rate, how far does the cyclist ride in 6 hours? _____ miles



2. A ratio graph of trays (x) and muffins (y) passes through $(1, 12)$. Find each missing value.

Step 1 The unit rate is 12 muffins per tray.

Step 2 To find y , multiply x by 12. To find x , divide y by 12.

a. $x = 4$, $y =$ _____

b. $y = 60$, $x =$ _____

c. $x = 7$, $y =$ _____

d. $y = 36$, $x =$ _____

e. $x = 10$, $y =$ _____

f. $y = 96$, $x =$ _____

- 3. Same Line?** A ratio graph of pounds of grapes (x) and cost in dollars (y) passes through (4, 10).

Step 1 Divide 10 by 4 to find the unit rate.

Step 2 For each point, divide y by x. Does it match?

- a.** What is the unit rate? _____ per pound
- b.** Is (6, 15) on the line? Is (8, 18)? Show how you know. _____

- 4. Find the Error.** A graph of laps (x) and minutes (y) passes through (3, 12). Omar says it means 12 laps in 3 minutes. What is wrong?

Frame Omar is wrong because x stands for __, so (3, 12) means __.

TEST PRACTICE

choose the best answer

- 5.** A ratio graph passes through (5, 40). Which point is also on the line?
- A** (1, 8) **B** (6, 41) **C** (8, 5) **D** (10, 45)
- 6.** A graph of hours (x) and dollars earned (y) passes through (2, 18). What does (1, 9) mean?
- A** \$9 per hour **B** 9 hours for \$1 **C** \$18 in 1 hour **D** \$2 per hour

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Tell someone at home what (1, 9) means on a graph of hours and dollars.